IMPACT OF *CYP2D6* FUNCTIONAL ALLELIC VARIATIONS ON PHENOCONVERSION AND DRUG-DRUG INTERACTIONS

F. Storelli[1,2], A. Matthey[1*], S. Lenglet[3*], A. Thomas[3,4,5], J. Desmeules[1,2,4,6], Y. Daali[1,2,4,6]

[1] Division of Clinical Pharmacology and Toxicology, Geneva University Hospitals, Geneva, Switzerland

[2] Geneva-Lausanne School of Pharmacy, University of Geneva, Geneva, Switzerland

[3] Unit of Toxicology, CURML, Lausanne-Geneva, Switzerland

[4] Swiss Center for Applied Human Toxicology, Geneva, Switzerland

[5] Faculty of Biology and Medicine, Lausanne University Hospital, University of Lausanne, Lausanne, Switzerland

[6] Faculty of Medicine, University of Geneva, Geneva, Switzerland

* These authors have contributed equally to this work.

Corresponding author:

Dr Youssef Daali

Division of Clinical Pharmacology and Toxicology

Geneva University Hospitals

Tel: +41 22 379 54 30

Fax: +41 22 372 99 45

Email: youssef.daali@hcuge.ch

Number of words: 4023/4000

Number of tables and figures: 7/7

Number of references: 41/50

Key words: CYP2D6, pharmacogenetics, genotype-phenotype translation, phenoconversion, activity score, heterozygous, drug-drug interactions

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process which may lead to differences between this version and the Version of Record. Please cite this article as an 'Accepted Article', doi: 10.1002/cpt.889

This article is protected by copyright. All rights reserved.

ABSTRACT

We investigated whether CYP2D6 extensive metabolizers (EM) carrying a non-functional allele are at higher risk of phenoconversion to poor metabolizers in presence of CYP2D6 inhibitors. Seventeen homozygous carriers of two fully-functional alleles and seventeen heterozygous carriers of one fully-functional and one non-functional allele participated in this trial. Dextromethorphan 5mg and tramadol 10mg were given at each of the three study sessions. CYP2D6 was inhibited by duloxetine 60mg (session 2) and paroxetine 20mg (session 3). Higher rate of phenoconversion to IM with duloxetine (71% *versus* 25%, p=0.009) and to PM with paroxetine (94% *versus* 56%, p=0.011) was observed in heterozygous than homozygous EMs. The magnitude of drug-drug interaction (DDI) between dextromethorphan and paroxetine was higher in homozygous than in heterozygous subjects (14.6 versus 8.5, p<0.028). Our study suggests that genetic EMs may not represent a homogenous population and that available genetic data should be considered when addressing DDIs in clinical practice.

Accepted Article

This article is protected by copyright. All rights reserved.

INTRODUCTION

CYP2D6 isoenzyme is involved in the metabolism of around 20% of xenobiotics, including opioid analgesics, cardiovascular and antipsychotics drugs (1). A large variability in both its activity and expression has been observed in the general population and four main phenotypes have been described so far: ultrarapid metabolizers (UM) showing fast metabolic capacity, extensive metabolizers (EM) with normal activity, intermediate metabolizers (IM) with reduced enzymatic function and finally poor metabolizers (PM) lacking enzymatic activity. The frequencies of these four phenotypes vary according to ethnicity, indicating a genetic component of CYP2D6 variability (2, 3).

More than one hundred alleles of *CYP2D6* have currently been identified and described in the Human CYP Allele Nomenclature Database (4). Variant alleles may lead to complete, reduced or increased enzymatic function. CYP2D6 phenotype can be predicted from genotype data through an activity score which has been proposed by Gaedigk and colleagues. In this system a value is assigned to each allele: 0 for non-functional alleles, 0.5 for reduced-function alleles, 1 for fully-functional alleles and 2 for duplications of fully-functional alleles (5). The genetic-predicted activity score (gAS) is calculated by summing the individual values of the two *CYP2D6* alleles. The Clinical Pharmacogenetics Implementation Consortium (CPIC) has proposed the following phenotype predictions: , UMs are predicted from a gAS > 2, EMs are predicted from a gAS between 1 and 2, IMs are predicted from a gAS of 0.5 and PMs are predicted from a gAS of 0 (6). Based on these predictions, therapeutic recommendations for CYP2D6 substrates have been published and regularly updated (6-10). It should be noted that there is no actual consensus on translating *CYP2D6* genetic data into phenotype and other systems of classifications such as the one of the Dutch Pharmacogenetics Working Group (DPWG) may predict CYP2D6 phenotype differently for a same genotype (11).

Additionally, CYP2D6 is involved in many pharmacokinetic drug-drug interactions (DDIs), especially with antidepressants (12). Many studies have highlighted the impact of genetic polymorphism on the extent and severity of DDIs (13-20). Phenoconversion is used to define the phenomenon by which an enzymatic phenotype is modified into another by an environmental factor such as DDI or concomitant illness (21, 22).

CYP2D6 EM phenotype frequency is about 60-70% among Caucasians (21, 23) and is predicted from different allelic combinations, including two fully-functional alleles (gAS=2) and one fully-functional

This article is protected by copyright. All rights reserved.

with one non-functional allele (gAS=1), according to the CPIC classification (6). The question has been raised whether individuals carrying a non-functional allele combined with a functional allele (about 25% of Caucasians) might be at particular risk of phenoconversion compared to homozygous EM when enzymatic inhibitors are given concomitantly (24). In addition to this, the magnitude of DDIs has not been studied in prospective pharmacokinetic clinical trials in this particular subset of subjects. Therefore, the primary objective of the present study is to prospectively assess the risk of CYP2D6 phenoconversion in genetic-predicted EM (gEM) healthy volunteers when administered either a moderate (duloxetine) or strong (paroxetine) inhibitor of CYP2D6. The population of the present study is restricted to either carriers of two-fully functional *CYP2D6* alleles (gAS=2, further named gAS2) or carriers of one fully-functional and one non-functional *CYP2D6* allele (gAS=1, further named gAS1) because of their high frequency. Pharmacokinetics parameters of the two CYP2D6 substrates tramadol and dextromethorphan and their respective metabolites O-desmethyltramadol (M1) and dextrorphan are also characterized in the two groups in presence of duloxetine and paroxetine.

RESULTS

Study subjects

Among the 90 subjects included in the trial after given informed consent, 35 eligible healthy volunteers, mainly Caucasians (see **Table 1**), participated in the study and 34 volunteers completed it. One of the study participants was a PM at session 1 (dextromethorphan/dextrorphan urinary metabolic ratio ($UMR_{DEM/DOR}$) > 0.3) and was excluded from analysis following a Dixon Q test, with no implication on the significance of the primary endpoint.

Safety

Adverse events were systematically collected through self-reporting and followed-up until resolution. Adverse drug reactions were mainly observed with duloxetine regimen. They were all mild and included nauseas, vomiting, sleeping troubles, headaches, dizziness, mydriasis, diarrheas and tremor, in accordance with side effects reported by the manufacturers.

CYP2D6 phenotyping in urine

This article is protected by copyright. All rights reserved.

CYP2D6 UMR results are presented in table 2 and show significantly higher proportion of IM and PM phenotypes with duloxetine and paroxetine respectively in gAS1 compared to gAS2. The inhibitory effect of duloxetine regimen was modest when significant: $UMR_{DEM/DOR}$ increased by 1.57-fold ($CI_{95\%}$ 1.10-2.25, p=0.004) in gAS1 and of 1.82-fold ($CI_{95\%}$ 0.99-3.35, p=0.163) in gAS2 subjects. With paroxetine, this increase was higher in the two groups (37-fold).

Dextromethorphan and dextrorphan pharmacokinetics

Pharmacokinetic profiles of dextromethorphan and its metabolite dextrorphan are presented in **Figure 2** and **Table 3**, respectively. At each of the three sessions, dextromethorphan Area Under the Concentration-Time Curve (AUC) was significantly higher in gAS1 subjects compared to gAS2. The difference of mean AUC values between the two groups was reduced with CYP2D6 inhibition: it was 3.9-fold at baseline, 2.9-fold at session 2 and 2.2-fold at session 3. Dextromethorphan maximal concentration (C_{max}) was higher in gAS1 subjects compared to gAS2 at baseline and the difference between the two groups tended to decrease when inhibiting CYP2D6: 2.7-fold at baseline, 2.6 fold at session 2 and 1.3-fold (non significant) at session 3.

Tramadol and O-desmethyltramadol pharmacokinetics

Pharmacokinetic profiles and parameters of tramadol and active metabolite M1 are presented in **Figure 2** and **Table 4**, respectively. Tramadol AUC and Cmax were weakly increased after the administration of duloxetine and paroxetine. The AUC and Cmax values of M1 in the three study sessions were respectively about 30-40% and 60% higher in gAS2 than in gAS1 subjects. With paroxetine, AUC and C_{max} of M1 decreased by about 60% in the two groups (p<0.001).

This article is protected by copyright. All rights reserved.

Accepted Article

DISCUSSION

Phenoconversion

Although baseline phenotype distribution was not significantly different between homozygous and heterozygous EMs, a tendency was noted towards higher IM frequency among these latter. 41% of gAS1 subjects were phenotypically IM at baseline and the mean $UMR_{DEM/DOR}$ value in this group was equal to the accepted cut-off value to discriminate between EM and IM phenotypes. Thus, one can ask itself as whether it would be relevant to classify heterozygous gAS1 subjects as either EMs or IMs. There is currently no universally accepted method for CYP2D6 phenotype assignment based on genetic data and some discrepancies exist between genotype-phenotype translation by the CPIC, the DPWG and different laboratories. This leads to variable interpretations and therefore different therapeutic measures for a same genotype. While a heterozygous gAS1 genotype is translated into an EM phenotype according to the CPIC classification, the DPWG classification predicts an IM phenotype (11). The 4-group phenotype classification has traditionally been used, with the major advantage of being intuitive and easily comprehensible for application in clinical practice. However, *CYP2D6* diplotypes represent a continuum of activity and this simple classification might be responsible for loss of information. A 7-group classification using the activity score system has been proposed by some authors in order to take into account the continuum of activity (11). To our knowledge this has not been adopted yet, probably because of its complexity making it difficult to implement. The 5-group phenotype classification discriminating slow (gAS1) and fast EMs (gAS 1.5-2) (3) might represent a compromise, being more intuitive than the 7-group classification while addressing the variability of activity in the EM population.

When administered with duloxetine, dextromethorphan AUC increased 1.5-3.2-fold , which confirms moderate CYP inhibition . None of the study subjects in either group was phenoconverted into PM with duloxetine. Duloxetine mean plasma maximal concentration in our study was about 100 ng/ml (**Figure 3**), which amounts to the one reported for a 60 mg twice-daily regimen at steady-state (25). The significant 2.8-fold difference in IM phenotype proportion between gAS1 and gAS2 subjects might be explained by the baseline IM proportion. As shown in **Figure 1**, the impact of duloxetine on $UMR_{DEM/DOR}$ was moderate. Because baseline $UMR_{DEM/DOR}$ values in gAS1 subjects were close to the EM-IM cut-off value, a mean increase of 1.9-fold was sufficient to get 71% of gAS1 subjects IM with

This article is protected by copyright. All rights reserved.

duloxetine. In both study groups, IM proportion was increased by about 2-fold when co-medicated with duloxetine.

Paroxetine is a strong inhibitor of CYP2D6 , it increases the AUC of metoprolol, desipramine and atomoxetine by more than 5-fold (26). In this study, two close 20 mg doses were given before the administration of probe drugs, so that paroxetine maximal plasma levels were approximately the same as those reached with a 20 mg daily regimen at steady-state (between 12 and 90 ng/ml with a mean of 40 ng/ml) (27). Because paroxetine is metabolized by CYP2D6, its plasma exposure was higher in gAS1 subjects than gAS2 (**Figure 3**), which also explains the higher rate of phenoconversion in the gAS1 group. The rate of phenoconversion among gAS1 subjects was massive when inhibited by paroxetine, for all subjects except one becoming PM. This rate was significantly lower among gAS2 subjects: only about half of them turned to PM.

The present observations suggest that genotyping might represent a fast, cost-limited and powerful method of identifying patients at higher risk of phenoconversion in presence of concomitant CYP inhibitors. Those patients might include genetic IM carrying one non-functional and one reduced-function allele and genetic EM carrying a non-functional allele. Whether genetic EM carrying one or two reduced-function alleles might be included in this high-risk subset needs to be established in further clinical studies.

Because CYP2D6 activity is influenced by several intrinsic and extrinsic factors, the prediction of phenotype based only on genetic data is unsatisfactory. Borges and colleagues adapted the activity score system developed by Gaedigk et al. (5) by incorporating the impact of concomitant medications in the activity score calculation (28). The integration of genotype independent factors in the composite activity score calculation nearly doubled the ability to predict CYP2D6 phenotype. Using this system with our data, a Pearson association of R^2=0.61 ($p<0.001$) was found between the logarithm of $UMR_{DEM/DOR}$ and the composite gene-DDI activity score (**Supplementary figure S1**). Although such R^2 indicates strong association, it still means that about 40% of CYP2D6 phenotype, as measured by $UMR_{DEM/DOR}$, cannot be predicted from this composite genetic-DDI activity score. In our opinion, measuring the actual function of CYP2D6 before drug administration represents the most accurate method for predicting the actual exposure of a drug. Phenotyping takes into account endogenous, environmental as well as genetic factors and is usually performed by administrating a specific

This article is protected by copyright. All rights reserved.

substrate such as dextromethorphan or metoprolol. A metabolic ratio in either blood, plasma (29) or urine (30) serves as a metrics to classify subjects into four phenotypic groups. However, the major current limitation of phenotyping is the safety concerns related to drug administration, now minimized by the development of highly sensitive analytical methods allowing to use subtherapeutic doses (31). In addition, further research is required and reference values to classify subjects into phenotypic groups need to be validated in large cohorts before implementing CYP2D6 phenotyping in clinical practice.

Extent of Drug-drug interactions

The magnitude of DDIs (AUC_i/AUC) relies on several factors, which are functions of both the perpetrator and the victim drugs: enzymatic inhibition parameters of inhibitor, intracellular unbound concentration of inhibitor and contribution of the inhibited enzyme to the total clearance of the victim drug, as shown in static equations (1) and (2) for competitive and time-dependent inhibitions, respectively (32-34).

$$\frac{AUC_i}{AUC} = \frac{1}{\left[\frac{1}{1+\frac{[I]_h}{K_i}}\right] \cdot f_{m,CYP2D6} + \left(1 - f_{m,CYP2D6}\right)} \quad (1)$$

$$\frac{AUC_i}{AUC} = \frac{1}{\left[\frac{k_{deg,h}}{k_{deg,h} + \frac{[I]_h \cdot k_{inact}}{[I]_h + K_I}}\right] \cdot f_{m,CYP2D6} + \left(1 - f_{m,CYP2D6}\right)} \quad (2)$$

$[I]_h$ is the intracellular unbound liver concentration of inhibitor, $f_{m,CYP2D6}$ is the fraction of systemic clearance of the substrate mediated by CYP2D6. K_i is the constant of inhibition for competitive inhibition. For time-dependant inhibitors, k_{inact} and K_I are the maximal inactivation rate of enzyme and concentration of inhibitor at which half the k_{inact} value is reached, respectively.

The effect of non-functional allelic variants *3-8 of *CYP2D6* on the extent of DDIs is not expected to affect inhibition parameters K_i, K_I and k_{inact}, because the proteins resulting from theses mutations already lack of functionality and cannot be inhibited. Their impact relies then on the $f_{m,CYP2D6}$ value, which depends on the abundance of the enzyme and/or its functional activity. Therefore, $f_{m,CYP2D6}$ is expected to be lower in gAS1 subjects than in gAS2. Available $f_{m,CYP2D6}$ data for dextromethorphan and tramadol were 0.96 (major clearance pathway) and 0.31 (minor clearance pathway) (See

This article is protected by copyright. All rights reserved.

Accepted Article

Supplementary Information for more details on input data for static predictions of DDIs). The AUC ratio of dextromethorphan-paroxetine, dextromethorphan-duloxetine, tramadol-paroxetine and tramadol-duloxetine predicted from static equations (1) and (2) were 14.7, 2.7, 1.4 and 1.3, respectively. The difference in $f_{m,CYP2D6}$ values for dextromethorphan and tramadol explains the higher magnitudes of DDI for the first drug compared to the second. Static-predicted values are close to those observed in the present study for homozygous gAS2. Decreasing $f_{m,CYP2D6}$ values to mimic the impact of gAS1 genotype resulted in lower AUC ratios. Similar predicted and observed AUC ratio were found when attributing a $f_{m,\ CYP2D6}$ value of 0.91 and 0.25 for dextromethorphan and tramadol respectively: 8.7 for the interaction paroxetine-dextromethorphan, 2.5 for the interaction duloxetine-dextromethorphan, 1.3 for the interaction paroxetine-tramadol and 1.2 for the interaction duloxetine-tramadol. The mechanistic-static predictions are in accordance with our data and show that the higher the contribution of the inhibited pathway in total clearance of the victim drug, the higher is the extent of DDI.

Dextromethorphan exposure was always significantly higher in gAS1 subjects, whether CYP2D6 was inhibited or not. When CYP2D6 was inhibited by paroxetine, dextromethorphan exposure was strongly increased. Consequently, drug concentrations of CYP2D6 substrates might possibly reach the toxicity range in gAS1 subjects but not in gAS2, depending on the therapeutic windows. However, when a perpetrator inhibiting a major metabolic pathway is added after dose optimization of the victim drug through therapeutic drug monitoring, gAS2 subjects might be at higher risk of toxicity than gAS1. Indeed, the victim drug's baseline concentration (before addition of the inhibitor) is the same for gAS2 and gAS1 but the magnitude of DDI is higher for gAS2. Therefore, the latter tend to be exposed to higher drug concentrations, which might result in higher incidence of adverse drug reactions.

Tramadol elimination involves two main routes: N-desmethylation by CYP3A and CYP2B6 and O-desmethylation to active metabolite M1 by CYP2D6 (35, 36). Although clinically significant because of its role in the formation of active metabolite, CYP2D6 contribution in tramadol clearance is moderate (37), which explains the weak impact of CYP2D6 inhibition on tramadol exposure in our study. M1 exposure was lower in gAS1 subjects than in gAS2 subjects both at baseline and when moderately inhibited by duloxetine. With paroxetine M1 AUC tended to be lower in gAS1 subjects but the difference was not significant. Maximal concentrations of M1 were always lower in gAS1 subjects than in gAS2. Our data showed that when the formation of M1, which is responsible for the analgesic effect

This article is protected by copyright. All rights reserved.

Accepted Article

of the drug (38), is inhibited by a strong CYP2D6 inhibitor such as paroxetine, plasma concentrations decreased by about 50-60% compared to the baseline. In gAS1 subjects, lower concentrations are found. Therefore, when co-medicated with strong CYP2D6 inhibitors the latter might possibly be insufficiently treated with standard tramadol dosage and thus require additional doses in order to reach adequate pain relief.

To conclude, the present study reveals the higher vulnerability of heterozygous gAS1 subjects to phenoconversion and highlights the need to consider CYP2D6 genotype for the management of DDIs in clinical practice. When facing gene-DDI interactions, both baseline CYP phenotype and predicted magnitude of interaction have to be taken into account. The results of this study suggest that heterozygous gAS1 should not be considered as a low-risk population, especially in the presence of concomitant medications, and that it might be rational to genotype patients who are about to receive a prescription of a CYP2D6 inhibitor in order to detect those at higher risk of phenoconversion. Moreover, theoretical discussion on the basis of static mechanistic equation of DDI prediction and observed PK data from the present study show that the greater CYP activity resulting from genetic polymorphism, the greater the impact of an inhibitor of this same CYP on the clearance of a victim drug mainly cleared by the inhibited pathway will be. In the case of a victim metabolized by multiple pathways, genetic polymorphism affecting the function of a major clearance pathway will increase the contribution of the alternative minor pathway and therefore results in significant DDI when an inhibitor of the minor pathway is given concomitantly (20, 39, 40).

Future perspectives

The main limitation of the present study lies in the dosing regimen of the two CYP inhibitors. In order to avoid the risks related to ambulatory intake of drugs for participant's safety and compliance, pre-doses with reduced (10-hours) interval were chosen instead of multiples administrations. The results of this exploratory pharmacokinetic study inspire further confirmation in patients with various combinations of alleles taking CYP inhibitors at steady-state to assess the real risk of phenoconversion and its impact on drug safety and efficacy. Further prospective clinical trials studying the impact of other genetically-predicted phenotypes on the extent of DDIs and risk of phenoconversion are required.

This article is protected by copyright. All rights reserved.

Accepted Article

Physiologically-based Pharmacokinetic (PBPK) modeling has shown an ability to predict complex DDIs involving genetic polymorphism or race-related CYP abundance (39, 41) and has been proposed as an important bridging tool in drug development to address complex scenarios combining intrinsic and extrinsic factors. Some authors have recommended a framework for the prediction of such scenarios in drug development (39, 41). Preclinical and early clinical phases provide a lot of in vitro and PK data allowing for the elaboration of a PBPK model for new drug entity under development. Interactions and pharmacogenetic trials performed during clinical phases then provide valuable information for quantitative assignment of clearance pathways and recalibration of initial PBPK model. Provided that input in vitro and clinical data have a high degree of quality, complex scenarios of DDIs with a set of anticipated concomitant inhibitors could then be predicted in order to support dose adjustment recommendations. Further research is required and PBPK models to address gene-drug-drug interactions are currently under development.

METHODS

Study population

90 healthy men and women aged from 18 to 60 years were included in the trial and genotyped for *CYP2D6*. Those with a *CYP2D6* genotype composed of two fully-functional alleles (*1, *2, *35), gAS2, and those with one fully-functional and one non-functional allele (*3-8), gAS1, with body-mass index (BMI) 18-27 kg/m^2 were eligible to participate in the trial. The exclusion criteria were the following: pregnancy or breastfeeding, any pathology, drug or food affecting CYP activity, tobacco consumption >5 cigarettes/day, renal and hepatic impairment, medical history of chronic alcoholism or abuse of psychoactive drugs, liver transplantation and drug sensitivity.

The study protocol was reviewed and approved by Research Ethics Committee of Geneva as well as the Swiss Agency for Therapeutic Drugs (Swissmedic). All participants provided written informed consent prior to inclusion. Protocol conception and trial conduct were performed in accordance with the Declaration of Helsinki ethical principles and the Good Clinical Practice guidelines of the International Congress of Harmonization. The trial is registered at http:// www.clinicaltrials.gov (NCT03054220).

CYP2D6 genotyping

This article is protected by copyright. All rights reserved.

CYP2D6 genotyping from whole blood was performed by Sanger sequencing on PCR products. The following mutations were screened: *1 (wild-type, default allele), *2 (2850C>T), *3 (2549delA), *4 (1846G>A), *6 (1707delT), *7 (2935A>C), *8 (1758G>T), *9 (2615_2617delAAG), *10 (100C>T), *17 (1023C>T), *29 (1659G>A), *35 (31G>A), *41 (2988G>A). Gene deletion (*5 allele) and duplication were detected with Taqman® Copy Number Assay. Details are available **Supplementary Information**.

Study design

This parallel open-label study was performed in three sessions for all participants separated into two groups: gAS1 and gAS2. Wash-out periods of at least one and two weeks were respected between sessions 1 and 2 and sessions 2 and 3, respectively. At each of the three sessions 5 mg of dextromethorphan (BEXIN syrup®, Spirig Healthcare, Egerkingen, Switzerland) and 10 mg of tramadol (TRAMAL® drops, Grünenthal Pharma, Glarus, Switzerland) were orally administered concomitantly to study participants after an overnight fast. Venous blood samples were collected before and at 0.5, 1, 2, 3, 4, 6, 8, 10 and 24 hours after drug administration into tubes containing EDTA (BD Vacutainer, Plymouth, UK). Plasma was obtained though centrifugation at 2750 g for 10 minutes and stored at -20°C before analysis. In addition, participants were asked to collect their urine up to 10 hours after the drugs' administration. A 20 ml sample of urine was stored at -20°C until analysis. At session 2 the study course was similar but participants were asked to take 60 mg of duloxetine (CYMBALTA®, Eli Lilly, Vernier, Switzerland) 12 and 2 hours before dextromethorphan and tramadol administration. At session 3, participants were asked to take 20 mg of paroxetine (PAROXETIN-MEPHA®, Basel, Switzerland) 12 and 2 hours before dextromethorphan and tramadol administration. Participants were specifically asked about the time of duloxetine and paroxetine intakes and were requested to bring back empty blisters to verify compliance.As for the women participating in the study, a pregnancy test was performed at inclusion and at each session before any drug administration. Breakfast and lunch were given 1 and 4 hours after drug administration, respectively.

Quantification of drugs and metabolites levels in plasma and urine

This article is protected by copyright. All rights reserved.

All the drugs and metabolites were quantified in plasma by liquid chromatography coupled to an API4000 triple quadrupole mass spectrometer (Sciex, Darmstadt, Germany) (LC-MS/MS) following liquid-liquid extraction (LLE).

Deconjugated dextromethorphan and dextrorphan levels in urine were also quantified by LC-MS/MS following chemical hydrolysis and LLE. As previously described (23, 30), CYP2D6 phenotype was assigned based on $UMR_{DEM/DOR}$. PM phenotype was characterized by an $UMR_{DEM/DOR}$ >0.3, IM phenotype by an $UMR_{DEM/DOR}$ between 0.03 and 0.3, EM phenotype by an $UMR_{DEM/DOR}$ between 0.003 and 0.03 and UM phenotype by UMR<0.003. The analytical methods were fully-validated.

Data analysis and statistics

34 subjects (17 in each group) were required to have an 80% chance of detecting, as significant at the 5% level, an increase in the proportion of subjects phenoconverted to PM from 10% in gAS2 subjects to 50% in gAS1 subjects, *id est* a 40% difference in proportions.

Pharmacokinetic parameters were estimated by standard non-compartmental methods using WinNonlin version 6.2.1 (Pharsight, Mountain View, CA). Results are presented as geometric means with $CI_{95\%}$. Comparisons of phenotype proportions were performed by χ^2 statistical tests. Comparisons of pharmacokinetic parameters were performed by Wilcoxon non-parametric tests. Inter-sessions $UMR_{DEM/DOR}$ and AUC values were compared by a Wilcoxon non-parametric test for paired samples. All statistical analyses were performed using SPSS software version 22 (Chicago, IL, US). P-values ≤0.05 were considered statistically significant.

This article is protected by copyright. All rights reserved.

STUDY HIGHLIGHTS

What is the current knowledge on the topic?

Many allelic combinations of CYP2D6 predict the frequent extensive metabolizer (EM) phenotype. Genetic EMs carrying a non-functional allele are thought to be at higher risk of phenoconversion but no appropriate prospective clinical trial has addressed this question so far.

What question did this study address?

The study aimed to assess the risk of CYP2D6 phenoconversion and the magnitude of drug-drug interactions (DDIs) in heterozygous EMs carrying one non-functional allele and homozygous EMs carrying two fully-functional alleles.

What this study adds to our knowledge?

Although often predicted EMs, carriers of a non-functional allele have lower enzymatic activity and are at higher risk of CYP2D6 phenoconversion than homozygous EMs. The elimination and bioactivation of CYP2D6 substrates are also significantly impacted by the presence of a non-functional allele, as well as the magnitude of DDIs.

How this might change clinical pharmacology or translational science?

If available, genotype data need to be considered to improve DDIs prediction. If so, both baseline CYP phenotype and predicted magnitude of interaction have to be taken into account.

This article is protected by copyright. All rights reserved.

Accepted Article

ACKNOWLEDGMENTS

The authors would like to thank the Clinical Research Center (Geneva University Hospital and Faculty of Medicine) for helpful contribution, Antoine Poncet for statistical support and Mélanie Kuntzinger for technical assistance. This work was supported by the Department of Anesthesiology, Pharmacology and Intensive Care of Geneva University Hospitals.

CONFLICT OF INTEREST/DISCLOSURE

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Y.D., F.S., A.M., S.L., A.T., and J.D. wrote the manuscript; Y.D., F.S., A.M., and J.D. designed the research; F.S., A.M., and S.L. performed the research; Y.D., F.S., A.M., and J.D. analyzed the data; A.T. contributed new reagents/analytical tools.

Accepted Article

This article is protected by copyright. All rights reserved.

REFERENCES

(1) Zanger, U.M., Raimundo, S. & Eichelbaum, M. Cytochrome P450 2D6: overview and update on pharmacology, genetics, biochemistry. *Naunyn Schmiedebergs Arch Pharmacol* **369**, 23-37 (2004).

(2) Ingelman-Sundberg, M. Genetic polymorphisms of cytochrome P450 2D6 (CYP2D6): clinical consequences, evolutionary aspects and functional diversity. *Pharmacogenomics J* **5**, 6-13 (2005).

(3) Gaedigk, A., Sangkuhl, K., Whirl-Carrillo, M., Klein, T. & Leeder, J.S. Prediction of CYP2D6 phenotype from genotype across world populations. *Genet Med* **19**, 69-76 (2017).

(4) Human Cytochrome P450 (CYP) Allele Nomenclature Database. Available from: http://www.cypalleles.ki.se.

(5) Gaedigk, A., Simon, S.D., Pearce, R.E., Bradford, L.D., Kennedy, M.J. & Leeder, J.S. The CYP2D6 activity score: translating genotype information into a qualitative measure of phenotype. *Clin Pharmacol Ther* **83**, 234-42 (2008).

(6) Crews, K.R. *et al.* Clinical Pharmacogenetics Implementation Consortium guidelines for cytochrome P450 2D6 genotype and codeine therapy: 2014 update. *Clin Pharmacol Ther* **95**, 376-82 (2014).

(7) Hicks, J.K. *et al.* Clinical Pharmacogenetics Implementation Consortium (CPIC) Guideline for CYP2D6 and CYP2C19 Genotypes and Dosing of Selective Serotonin Reuptake Inhibitors. *Clin Pharmacol Ther* **98**, 127-34 (2015).

(8) Hicks, J.K. *et al.* Clinical Pharmacogenetics Implementation Consortium guideline for CYP2D6 and CYP2C19 genotypes and dosing of tricyclic antidepressants. *Clin Pharmacol Ther* **93**, 402-8 (2013).

(9) Bell, G.C. *et al.* Clinical Pharmacogenetics Implementation Consortium (CPIC) guideline for CYP2D6 genotype and use of ondansetron and tropisetron. *Clin Pharmacol Ther*, (2016).

(10) Hicks, J.K. *et al.* Clinical pharmacogenetics implementation consortium guideline (CPIC) for CYP2D6 and CYP2C19 genotypes and dosing of tricyclic antidepressants: 2016 update. *Clin Pharmacol Ther*, (2016).

(11) Hicks, J.K., Swen, J.J. & Gaedigk, A. Challenges in CYP2D6 phenotype assignment from genotype data: a critical assessment and call for standardization. *Curr Drug Metab* **15**, 218-32 (2014).

(12) Ereshefsky, L., Riesenman, C. & Lam, Y.W. Antidepressant drug interactions and the cytochrome P450 system. The role of cytochrome P450 2D6. *Clin Pharmacokinet* **29 Suppl 1**, 10-8; discussion 8-9 (1995).

(13) Lim, K.S. *et al.* Pharmacokinetic interaction of flecainide and paroxetine in relation to the CYP2D6*10 allele in healthy Korean subjects. *Br J Clin Pharmacol* **66**, 660-6 (2008).

(14) Suzuki, Y. *et al.* Importance of the cytochrome P450 2D6 genotype for the drug metabolic interaction between chlorpromazine and haloperidol. *Ther Drug Monit* **23**, 363-8 (2001).

(15) LLerena, A., Berecz, R., de la Rubia, A., Fernandez-Salguero, P. & Dorado, P. Effect of thioridazine dosage on the debrisoquine hydroxylation phenotype in psychiatric patients with different CYP2D6 genotypes. *Ther Drug Monit* **23**, 616-20 (2001).

(16) Madadi, P. *et al.* Fatal hydrocodone overdose in a child: pharmacogenetics and drug interactions. *Pediatrics* **126**, e986-9 (2010).

(17) Azuma, J. *et al.* The relationship between clinical pharmacokinetics of aripiprazole and CYP2D6 genetic polymorphism: effects of CYP enzyme inhibition by coadministration of paroxetine or fluvoxamine. *Eur J Clin Pharmacol* **68**, 29-37 (2012).

(18) Tod, M., Nkoud-Mongo, C. & Gueyffier, F. Impact of genetic polymorphism on drug-drug interactions mediated by cytochromes: a general approach. *AAPS J* **15**, 1242-52 (2013).

(19) Haffen, E. *et al.* Citalopram: an interaction study with clomipramine in a patient heterozygous for CYP2D6 genotype. *Pharmacopsychiatry* **32**, 232-4 (1999).

(20) Bahar, M.A., Setiawan, D., Hak, E. & Wilffert, B. Pharmacogenetics of drug-drug interaction and drug-drug-gene interaction: a systematic review on CYP2C9, CYP2C19 and CYP2D6. *Pharmacogenomics* **18**, 701-39 (2017).

(21) Zanger, U.M. & Schwab, M. Cytochrome P450 enzymes in drug metabolism: regulation of gene expression, enzyme activities, and impact of genetic variation. *Pharmacol Ther* **138**, 103-41 (2013).

(22) Shah, R.R. & Smith, R.L. Addressing phenoconversion: the Achilles' heel of personalized medicine. *Br J Clin Pharmacol* **79**, 222-40 (2015).

This article is protected by copyright. All rights reserved.

Accepted Article

(23) Rebsamen, M.C. *et al.* The AmpliChip CYP450 test: cytochrome P450 2D6 genotype assessment and phenotype prediction. *Pharmacogenomics J* **9**, 34-41 (2009).
(24) Berm, E.J., Risselada, A.J., Mulder, H., Hak, E. & Wilffert, B. Phenoconversion of cytochrome P450 2D6: the need for identifying the intermediate metabolizer genotype. *J Clin Psychiatry* **74**, 1025 (2013).
(25) Knadler, M.P., Lobo, E., Chappell, J. & Bergstrom, R. Duloxetine: clinical pharmacokinetics and drug interactions. *Clin Pharmacokinet* **50**, 281-94 (2011).
(26) Venkatakrishnan, K. & Obach, R.S. In vitro-in vivo extrapolation of CYP2D6 inactivation by paroxetine: prediction of nonstationary pharmacokinetics and drug interaction magnitude. *Drug Metab Dispos* **33**, 845-52 (2005).
(27) Deroxat(R) [package insert]. Münchenbuchsee, Switzerland: GlaxoSmithKline AG; 2015.
(28) Borges, S. *et al.* Composite functional genetic and comedication CYP2D6 activity score in predicting tamoxifen drug exposure among breast cancer patients. *J Clin Pharmacol* **50**, 450-8 (2010).
(29) Bosilkovska, M. *et al.* Geneva cocktail for cytochrome p450 and P-glycoprotein activity assessment using dried blood spots. *Clin Pharmacol Ther* **96**, 349-59 (2014).
(30) Daali, Y., Cherkaoui, S., Doffey-Lazeyras, F., Dayer, P. & Desmeules, J.A. Development and validation of a chemical hydrolysis method for dextromethorphan and dextrophan determination in urine samples: application to the assessment of CYP2D6 activity in fibromyalgia patients. *J Chromatogr B Analyt Technol Biomed Life Sci* **861**, 56-63 (2008).
(31) Bosilkovska, M. *et al.* Simultaneous LC-MS/MS quantification of P-glycoprotein and cytochrome P450 probe substrates and their metabolites in DBS and plasma. *Bioanalysis* **6**, 151-64 (2014).
(32) Food and Drug Administration. Guidance for Industry: Drug Interaction Studies - Study Design, Data Analysis, Implications for Dosing, and Labeling Recommendations. February 2012.
(33) Peters, S.A. *Physiologically-Based Pharmacokinetic (PBPK) Modeling and Simulations: Principles, Methods, and Applications in the Pharmaceutical Industry.* (John Wiley & Sons: Hoboken, New Jersey, USA, 2012).
(34) Rowland, M. & Matin, S.B. Kinetics of drug-drug interactions. *Journal of Pharmacokinetics and Biopharmaceutics* **1**, 553-67 (1973).
(35) Paar, W.D., Poche, S., Gerloff, J. & Dengler, H.J. Polymorphic CYP2D6 mediates O-demethylation of the opioid analgesic tramadol. *Eur J Clin Pharmacol* **53**, 235-9 (1997).
(36) T'Jollyn, H. *et al.* Strategies for Determining Correct Cytochrome P450 Contributions in Hepatic Clearance Predictions: In Vitro-In Vivo Extrapolation as Modelling Approach and Tramadol as Proof-of Concept Compound. *Eur J Drug Metab Pharmacokinet*, (2016).
(37) T'Jollyn, H. *et al.* Physiology-based IVIVE predictions of tramadol from in vitro metabolism data. *Pharm Res* **32**, 260-74 (2015).
(38) Lassen, D., Damkier, P. & Brosen, K. The Pharmacogenetics of Tramadol. *Clin Pharmacokinet* **54**, 825-36 (2015).
(39) Vieira, M.D. *et al.* PBPK model describes the effects of comedication and genetic polymorphism on systemic exposure of drugs that undergo multiple clearance pathways. *Clin Pharmacol Ther* **95**, 550-7 (2014).
(40) Mukai, Y. *et al.* Co-administration of Fluvastatin and CYP3A4 and CYP2C8 Inhibitors May Increase the Exposure to Fluvastatin in Carriers of CYP2C9 Genetic Variants. *Biol Pharm Bull* **40**, 1078-85 (2017).
(41) Patel, C., Rathi, C. & Venkatakrishnan, K. Should Race-Genotype Interactions Be Considered in the Global Development of CYP2C19 Substrates? A Proposed Framework Using Physiologically Based Pharmacokinetic Modeling. *J Clin Pharmacol* **57**, 417-21 (2017).

Accepted Article

This article is protected by copyright. All rights reserved.

FIGURE LEGENDS

Figure 1. 10-hour dextromethorphan/dextrorphan individual urinary metabolic ratios ($UMR_{DEM/DOR}$) at baseline (session 1), when inhibited by (A) duloxetine (session 2) and (B) paroxetine (session 3) following intake of dextromethorphan 5 mg per os. Homozygous carriers of two fully-functional alleles of CYP2D6 (gAS2) are shown in red with triangle markers and continuous lines. Heterozygous carriers of one fully-functional and one non-functional allele (gAS1) are shown in blue with diamond markers and dashed lines. Error bars represent the standard deviations of log-transformed $UMR_{DEM/DOR}$. All P-values were < 0.01.

Figure 2. Plasma concentration-time profiles of (A) dextromethorphan 5 mg per os, (B) tramadol 10 mg per os and their respective metabolites (C) dextrorphan and (D) O-desmethyltramadol (M1). Blue triangles represent drug profiles at session 1 (control), green diamonds represent drug profiles at session 2 (inhibition with duloxetine) and red spots represent drug profiles at session 3 (inhibition with paroxetine). Dashed and continuous lines stand for heterozygous extensive metabolizers carrying one non-functional allele (gAS1) and homozygous carriers of two fully-functional alleles (gAS2) subjects, respectively. Error bars represent standard deviations.

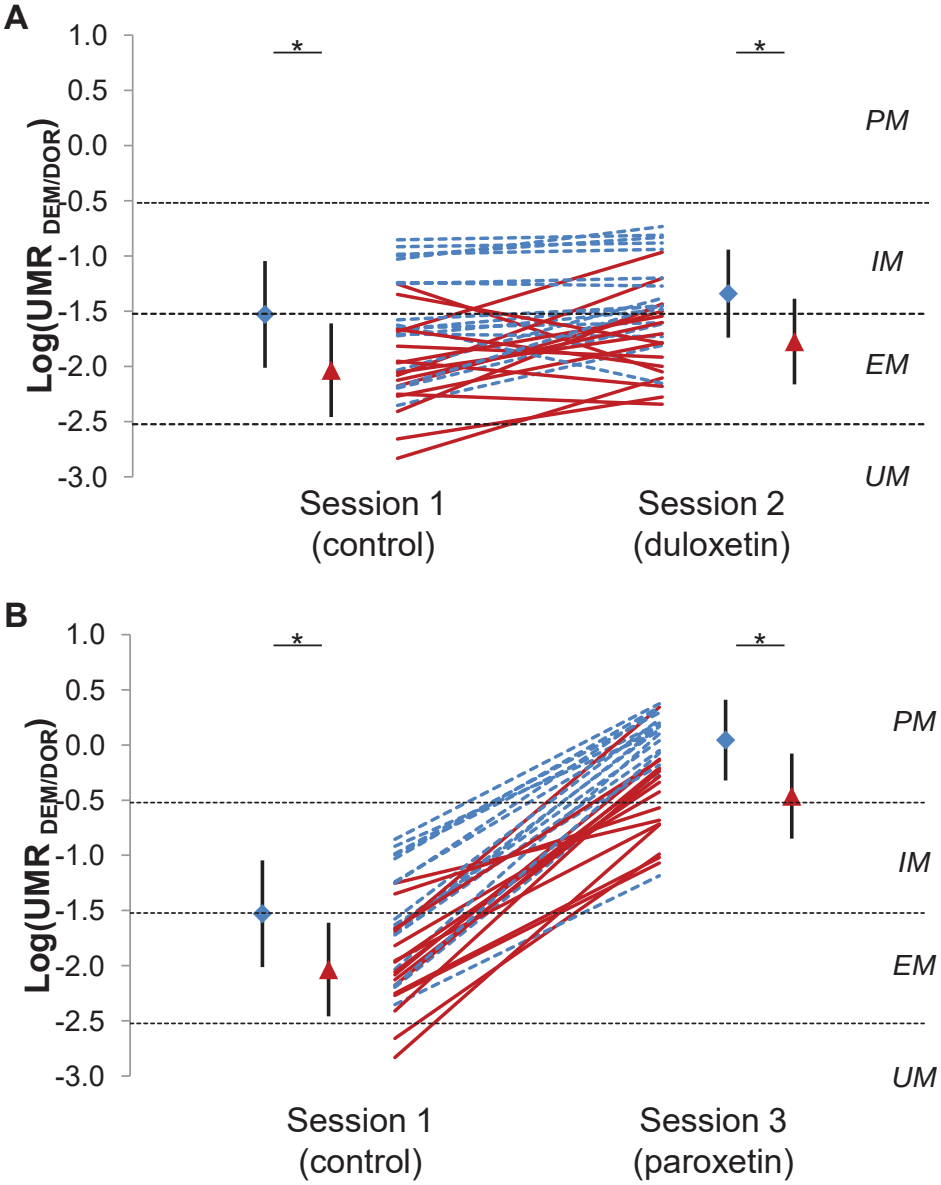
Figure 3. Plasma concentration-time profiles of (A) duloxetine and (B) paroxetine. Duloxetine 60 mg and paroxetine 20 mg were given before session 2 and 3 respectively 12 hours and 2 hours before administration of dextromethorphan 5 mg and tramadol 10 mg in order to inhibit CYP2D6. Drug profiles of heterozygous extensive metabolizers carrying one non-functional allele (gAS1) or homozygous carriers of two fully-functional alleles (gAS2) are shown in dashed and continuous lines respectively. Errors bars represent standard deviations.

Supplementary Figure S1. Correlation between dextromethorphan/dextrorphan urinary metabolic ratio ($UMR_{DEM/DOR}$) (log values) and the composite CYP2D6 activity score (AS) taking into account both genotype and enzymatic inhibition, as described by Borges and colleagues (28)

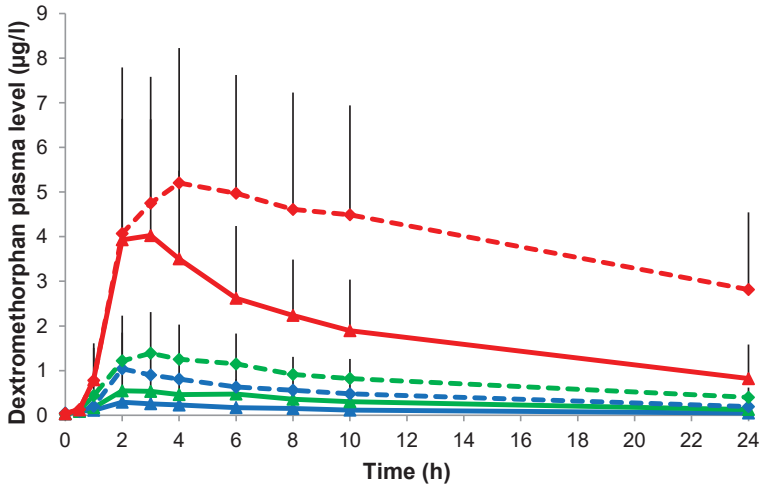
Supplementary Information

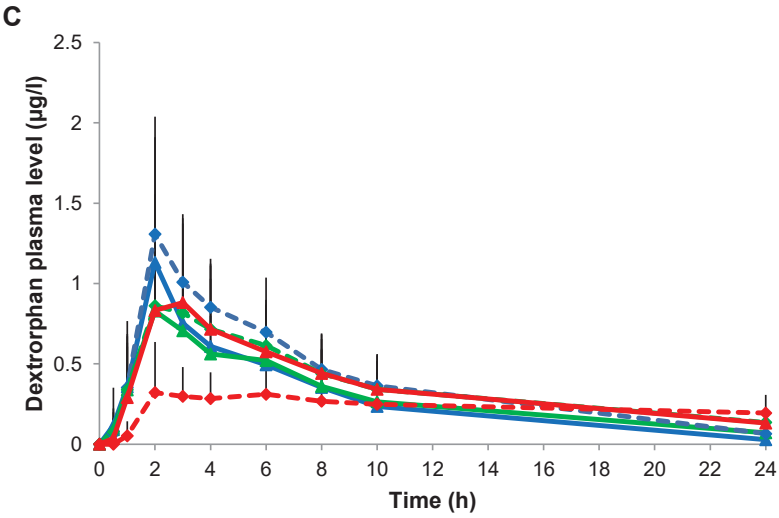
This article is protected by copyright. All rights reserved.

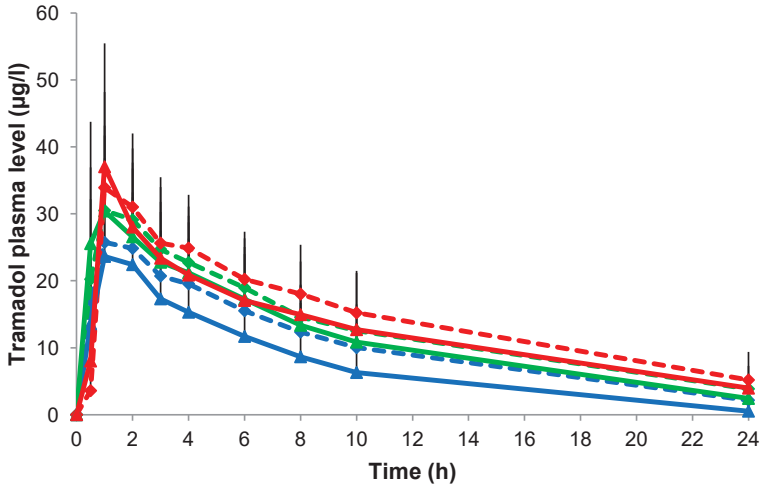
Accepted Article

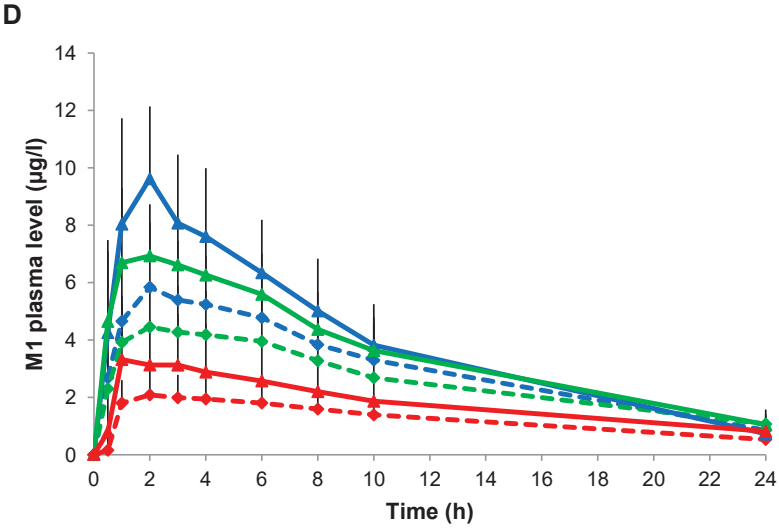


This article is protected by copyright. All rights reserved.

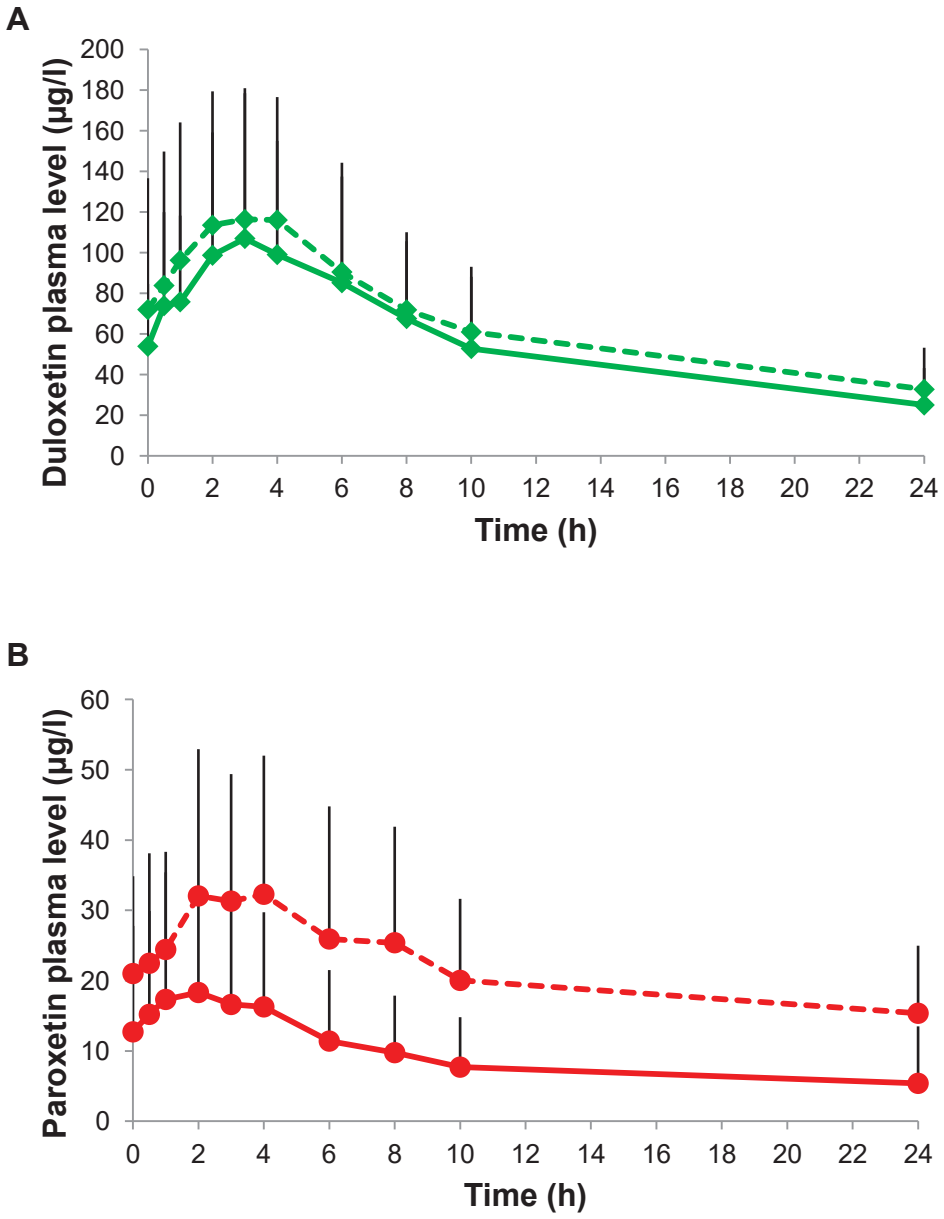








This article is protected by copyright. All rights reserved.



This article is protected by copyright. All rights reserved.

Table 1. Demographic characteristics of study subjects included in the analysis.

Parameter	**gAS1** *(n=17)*	**gAS2** *(n=16)*	**P-value**
Age (years)[(a)]	27 (18-42)	24 (21-27)	0.457[(b)]
BMI (kg/m^2)[(a)]	23 (19-26)	22 (19-26)	0.636[(b)]
Female (n)	9	12	0.188[(b)]
Ethnicity (n)			0.396[(c)]
Caucasian	15	15	-
Hispanic	-	1	-
Middle-East	1	-	-
Indian	1	-	-
***CYP2D6* allele frequencies**			
*1 (wild-type)	0.29	0.66	-
*2	0.06	0.28	-
*3	0.03	-	-
*4	0.32	-	-
*5	0.12	-	-
*6	0.03	-	-
*7	-	-	-
*8	-	-	-
*35	0.15	0.06	-

BMI, body-mass index.

[(a)] mean with minimal-maximal range

[(b)] Bilateral Mann-Whitney test

[(c)] Chi-square test

Accepted

This article is protected by copyright. All rights reserved.

Table 2. CYP2D6 phenotype distribution and urinary metabolic ratios (UMR) of heterozygous (gAS1) and homozygous (gAS2) extensive metabolizers at baseline and when inhibited with moderate (duloxetin) and strong (paroxetin) CYP2D6 inhibitors

		Session 1 (control)		**Session 2 (duloxetin)**		**Session 3 (paroxetin)**	
		gAS1 (n=17)	gAS2 (n=16)	gAS1 (n=17)	gAS2 (n=16)	gAS1 (n=17)	gAS2 (n=16)
Phenotype distribution[a]	**UM**	-	2 (12.5%)	-	-	-	-
	EM	10 (59%)	12 (75%)	5 (29%)	12 (75%)	-	-
	IM	7 (41%)	2 (12.5%)	12 (71%)	4 (25%)	1 (6%)	7 (44%)
	PM	-	-	-	-	16 (94%)	9 (56%)
	P-value (Chi-square)	0.085		0.009		0.011	
UMR[b]		0.030 (0.017-0.051)	0.009 (0.006-0.015)	0.046 (0.029-0.072)	0.017 (0.011-0.026)	1.106 (0.733-1.671)	0.344 (0.220-0.539)
P-value (Mann-Whitney)		0.006		0.005		<0.001	

UM, ultrarapid metabolizers; EM, extensive metabolizers; IM, intermediate metabolizers; PM, poor metabolizers

[a] Data are presented as number of subjects with proportion in percent.

[b] Data are presented as geometric means with 95% confidence intervals.

Accepted Article

This article is protected by copyright. All rights reserved.

Table 3. Pharmacokinetic parameters of dextromethorphan (DEM) and its metabolite dextrorphan (DOR) in heterozygous extensive metabolizers carrying one non-functional allele (gAS1) and homozygous carriers of two fully-functional alleles (gAS2) after administration of dextromethorphan 5 mg per os

Parameter	gAS1	gAS2	p-value
Dextromethorphan			
Session 1			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	8.7 (6.4-12.0)	2.3 (1.7-3.1)	<0.001
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	0.8 (0.6-1.2)	0.3 (0.2-0.4)	<0.001
Session 2			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	15.6 (11.7-20.9)	5.5 (3.8-8.0)	<0.001
AUC ratio	1.8 (1.5-2.1)	2.4 (1.8-3.2)	0.171
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	1.3 (0.9-1.7)	0.5 (0.3-0.7)	0.020
C_{max} ratio	1.5 (1.2-1.8)	1.9 (1.4-2.7)	0.264
Session 3			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	74.2 (53.3-103.3)	33.0 (21.7-50.3)	0.020
AUC ratio	8.5 (6.7-10.8)	14.6 (10.0-21.4)	0.028
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	4.8 (3.5-6.5)	3.7 (2.7-5.2)	0.368
C_{max} ratio	5.6 (4.1-7.7)	14.3 (10.4-19.6)	<0.001
Dextrorphan			
Session 1			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	8.9 (7.1-11.1)	5.4 (3.9-7.5)	0.013
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	1.2 (0.9-1.5)	1.0 (0.7-1.3)	0.113
Session 2			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	8.0 (6.3-10.2)	5.9 (4.3-8.2)	0.150
AUC ratio	0.9 (0.8-1.0)	1.1 (0.9-1.3)	0.132
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	0.8 (0.7-1.1)	0.8 (0.6-1.0)	0.407
C_{max} ratio	0.7 (0.6-0.9)	0.8 (0.6-1.0)	0.773
Session 3			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	4.8 (3.6-6.5)	7.6 (6.0-9.5)	0.048
AUC ratio	0.5 (0.4-0.7)	1.4 (1.1-1.9)	<0.001
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	0.3 (0.2-0.5)	0.9 (0.6-1.1)	<0.001
C_{max} ratio	0.3 (0.2-0.3)	0.9 (0.6-1.3)	<0.001
Metabolic ratio			
Session 1			
AUC_{DOR}/AUC_{DEM}	1.0 (0.7-1.4)	2.4 (1.8-3.1)	0.192
Session 2			
AUC_{DOR}/AUC_{DEM}	0.5 (0.4-0.7)	1.1 (0.8-1.4)	0.001
Fold decrease	2.0 (1.7-2.4)	2.2 (1.8-2.8)	0.719
Session 3			
AUC_{DOR}/AUC_{DEM}	0.1 (0.0-0.1)	0.2 (0.2-0.3)	<0.001
Fold decrease	15.6 (11.0-22.0)	10.4 (7.1-15.2)	0.130

AUC, Area Under the Concentration-Time Curve; C_{max}, maximal concentration.

All data are presented as geometric means with 95% confidence intervals.

This article is protected by copyright. All rights reserved.

Table 4. Pharmacokinetic parameters of tramadol and O-desmethyltramadol (M1) in heterozygous extensive metabolizers carrying one non-functional allele (gAS1) and homozygous carriers of two fully-functional alleles (gAS2) after administration of tramadol 10 mg per os

Parameter	gAS1	gAS2	p-value
Tramadol			
Session 1			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	233.5 (190.7-285.9)	152.7 (113.6-205.3)	0.023
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	27.5 (22.7-33.4)	23.6 (18.8-29.8)	0.280
Session 2			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	288.5 (237.2-351.0)	256.8 (204.9-321.8)	0.471
AUC ratio	1.2 (1.1-1.4)	1.7 (1.4-2.1)	0.014
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	31.8 (26.4-38.3)	31.0 (25.5-37.7)	0.943
C_{max} ratio	1.2 (1.0-1.3)	1.3 (1.1-1.5)	0.313
Session 3			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	324.3 (263.5-399.2)	255.9 (193.9-337.8)	0.140
AUC ratio	1.4 (1.2-1.6)	1.7 (1.2-2.4)	0.368
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	34.0 (28.0-41.2)	34.8 (28.8-42.1)	0.746
C_{max} ratio	1.2 (1.1-1.4)	1.5 (1.2-1.9)	0.407
M1			
Session 1			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	70.3 (59.6-83.1)	89.9 (74.3-108.9)	0.017
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	5.9 (4.9-7.2)	9.8 (8.4-11.5)	0.001
Session 2			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	59.2 (50.7-69.2)	83.9 (73.9-95.2)	0.027
AUC ratio	0.8 (0.8-0.9)	0.9 (0.8-1.1)	0.517
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	4.6 (3.9-5.5)	7.4 (6.6-8.3)	<0.001
C_{max} ratio	0.8 (0.7-0.9)	0.8 (0.7-0.9)	0.368
Session 3			
AUC ($mg \cdot h \cdot l^{-1} \times 10^{-3}$)	28.4 (23.7-34.0)	38.4 (31.2-47.3)	0.072
AUC ratio	0.4 (0.4-0.5)	0.4 (0.3-0.6)	0.971
C_{max} ($mg \cdot l^{-1} \times 10^{-3}$)	2.1 (1.7-2.6)	3.4 (2.7-4.2)	0.007
C_{max} ratio	0.4 (0.3-0.4)	0.3 (0.3-0.5)	0.746
Metabolic ratio M1/tramadol			
Session 1			
$AUC_{M1}/AUC_{Tramadol}$	0.3 (0.2-0.4)	0.6 (0.5-0.7)	<0.001
Session 2			
$AUC_{M1}/AUC_{Tramadol}$	0.2 (0.2-0.3)	0.3 (0.3-0.4)	0.001
Fold decrease	1.5 (1.3-1.6)	1.8 (1.5-2.1)	0.084
Session 3			
$AUC_{M1}/AUC_{Tramadol}$	0.1 (0.1-0.1)	0.2 (0.1-0.2)	0.006
Fold decrease	3.4 (2.9-4.1)	3.9 (3.3-4.7)	0.171

AUC, Area Under the Concentration-Time Curve; C_{max}, maximal concentration.

All data are presented as geometric means with 95% confidence intervals.

This article is protected by copyright. All rights reserved.